

Dynamic Guideline: WHO-Based Anaemia - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Most recent WHO anaemia-related guideline identified

- **Guideline Title:** *Guideline on haemoglobin cutoffs to define anaemia in individuals and populations*
- **Publication Date:** 5 March 2024
- **Issuing Body:** World Health Organization
- **Target Population:** Individuals and populations across life stages, including infants, children, adolescents, adults, pregnant women, and public-health populations.
- **Scope and Objectives:** To provide updated, clear, evidence-informed normative statements on haemoglobin concentrations for defining anaemia and on approaches to measuring anaemia in individuals and populations [WHO](#).
- **Intended Users:** Policy-makers, programme staff, clinicians, professional societies, nutrition and health programme managers, blood services, and public-health decision-makers [WHO](#).
- **Most important update:** The 2024 WHO guideline modified the haemoglobin cutoff for **children aged 6–23 months**, defining anaemia as haemoglobin **<105 g/L**; the previous broader 6–59 month cutoff was **<110 g/L** [Guideline Central](#), [WHO PDF](#).

Clarification on the requested youth-engagement title: The WHO item titled “*Meaningful youth engagement in a WHO guideline development process: case study of WHO Abortion care guideline (2022)*” is linked to the WHO abortion care guideline, not to anaemia. No WHO anaemia guideline or anaemia-specific case study with that title was identified in the supplied evidence. Therefore, it is **not counted** among the five WHO anaemia-related guidelines synthesized below [WHO event page](#).

1.2 Exactly 5 WHO Guidelines Related to Anaemia Synthesized

No.	WHO guideline or WHO normative guidance	Main anaemia relevance	Population
1	<i>Guideline on haemoglobin cutoffs to define anaemia in individuals and populations</i>	Diagnostic haemoglobin cutoffs, adjustment for altitude/smoking, population prevalence classification	Individuals and populations
2	WHO guidance on daily iron and folic acid supplementation in pregnancy, reflected in WHO ELENA and WHO consolidated maternal guidance	Prevention of maternal anaemia and adverse birth outcomes	Pregnant women
3	WHO context-specific intermittent iron and folic acid supplementation in pregnancy	Alternative regimen when daily iron is not acceptable and anaemia prevalence is low	Pregnant women

No.	WHO guideline or WHO normative guidance	Main anaemia relevance	Population
4	<i>Guideline: daily iron supplementation in infants and children</i>	Public-health prevention of anaemia and iron deficiency	Infants and children
5	WHO intermittent iron and folic acid supplementation for menstruating adolescent girls and women	Prevention of anaemia and improvement of haemoglobin and iron status in high-prevalence settings	Menstruating adolescent girls and women

1.3 WHO Anaemia Burden and Clinical Context

WHO estimates that anaemia remains highly prevalent globally:

- **40%** of children aged **6–59 months**
- **37%** of pregnant women
- **30%** of women aged **15–49 years**

are anaemic worldwide [WHO anaemia topic page](#).

Anaemia is defined clinically as a reduction in red blood cell number or haemoglobin concentration below normal, reducing oxygen-carrying capacity and causing symptoms such as fatigue, weakness, dizziness, and shortness of breath [WHO anaemia topic page](#). Causes include iron deficiency, folate deficiency, vitamin B12 deficiency, vitamin A deficiency, malaria, parasitic infections, tuberculosis, HIV, inflammation, haemoglobinopathies, chronic kidney disease, and other chronic diseases [WHO anaemia topic page](#), [WHO infants/children guideline PDF](#).

1.4 WHO 2024 Haemoglobin Cutoffs to Define Anaemia

The 2024 WHO guideline summary lists the following cutoffs, based on the **5th percentile**, for defining anaemia [Guideline Central](#):

Population	Haemoglobin concentration defining anaemia
Children, 6–23 months	<105 g/L
Children, 24–59 months	<110 g/L
Children, 5–11 years	<115 g/L
Children, 12–14 years, non-pregnant girls	<120 g/L
Children, 12–14 years, boys	<120 g/L
Adults, 15–65 years, non-pregnant women	<120 g/L
Adults, 15–65 years, men	<130 g/L
Pregnancy, first trimester	<110 g/L
Pregnancy, second trimester	<105 g/L
Pregnancy, third trimester	<110 g/L

Additional diagnostic principles:

- Adjust haemoglobin for **elevation of place of residence** when diagnosing anaemia [Guideline Central](#).
- Adjust haemoglobin for **smoking** [Guideline Central](#).
- Do **not** adjust haemoglobin for infection or inflammation when diagnosing anaemia, although inflammation remains important when interpreting ferritin and iron status [Guideline Central](#).

1.5 Public-Health Significance of Anaemia Prevalence

WHO maintains the following population-level classification [Guideline Central](#):

Anaemia prevalence	Public-health significance
≤4.9%	Normal
5.0–19.9%	Mild
20.0–39.9%	Moderate
≥40.0%	Severe

1.6 Key WHO Recommendations With GRADE Ratings

GRADE note: Where WHO source summaries reported explicit GRADE certainty or recommendation strength, those are shown. Where the supplied summaries did not provide full GRADE tables, certainty was judged using standard GRADE domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias. These report-level ratings should be interpreted as evidence-synthesis judgments, not as replacement for full WHO evidence profiles.

Recommendation	WHO source	GRADE Evidence Quality	Strength
Use WHO 2024 haemoglobin cutoffs to define anaemia in individuals and populations, including <105 g/L for children aged 6–23 months .	WHO 2024 haemoglobin-cutoff guideline WHO , Guideline Central	Moderate	Strong
Adjust haemoglobin for altitude/elevation of residence when diagnosing anaemia.	WHO 2024 haemoglobin-cutoff guideline summary Guideline Central	Moderate	Strong
Adjust haemoglobin for smoking when diagnosing anaemia.	WHO 2024 haemoglobin-cutoff guideline summary Guideline Central	Low	Conditional

Recommendation	WHO source	GRADE Evidence Quality	Strength
Do not adjust haemoglobin for infection or inflammation when defining anaemia, but assess inflammation when interpreting ferritin and iron biomarkers.	WHO 2024 haemoglobin-cutoff guideline summary Guideline Central	Low	Conditional
Use full blood count testing to diagnose anaemia in pregnancy where available.	WHO consolidated maternal guidance WHO consolidated guideline PDF	Moderate	Strong
Where full blood count is unavailable, use on-site haemoglobin testing with a haemoglobinometer rather than haemoglobin colour scale.	WHO consolidated maternal guidance WHO consolidated guideline PDF	Low	Conditional
Give daily oral iron and folic acid supplementation in pregnancy: 30–60 mg elemental iron + 400 µg folic acid daily.	WHO ELENA / maternal guidance WHO ELENA , WHO consolidated guideline PDF	Moderate	Strong
Use intermittent oral iron and folic acid in pregnancy — 120 mg elemental iron + 2.8 mg folic acid once weekly — when daily iron is not acceptable because of side effects and anaemia prevalence among pregnant women is <20% .	WHO maternal guidance / WIFAS document WHO WIFAS PDF , WHO consolidated guideline PDF	Low	Conditional
Use intravenous iron rather than oral iron in pregnancy with iron-deficiency anaemia when oral iron cannot be used, is not tolerated, or rapid correction is clinically needed, with capacity to monitor for anaphylaxis.	WHO consolidated maternal guidance WHO consolidated guideline PDF	Moderate	Conditional
Provide daily iron supplementation as a public-health intervention in infants and children to prevent anaemia and iron deficiency in appropriate settings.	WHO infants/children guideline WHO infants/children guideline page , WHO infants/children guideline PDF	Moderate overall; High for iron-deficiency anaemia outcome in 6–23 months	Strong

Recommendation	WHO source	GRADE Evidence Quality	Strength
Provide intermittent iron and folic acid supplementation to menstruating adolescent girls and women in settings where anaemia prevalence among non-pregnant adolescent girls and women aged 15–49 years is $\geq 20\%$.	WHO WIFAS guidance WHO WIFAS PDF	Moderate	Strong

1.7 Guideline Development Methodology

WHO’s 2024 haemoglobin-cutoff guideline was developed according to the **WHO handbook for guideline development** [NCBI Bookshelf](#).

Evidence Review Process

The process included:

- Prioritization of questions and outcomes.
- Evidence retrieval, assessment, and synthesis.
- Commissioned analysis of haemoglobin distributions in healthy populations.
- Systematic reviews and narrative evidence syntheses.
- Use of GRADE methodology where possible to prepare evidence profiles [WHO PDF](#), [NCBI Bookshelf](#).

Consensus Method

WHO established:

- A WHO Steering Committee led by the Department of Nutrition and Food Safety.
- A Guideline Development Group.
- A systematic reviews team.
- Regional WHO advisers participating in guideline meetings [NCBI Bookshelf](#).

The update began with an international consultation and addressed six subtopics, including anaemia physiology, cutoffs for population groups, clinical and environmental contexts, laboratory and diagnostic considerations, and implementation [NCBI Bookshelf](#).

Conflict of Interest Management

The supplied summaries state that WHO followed WHO guideline-development procedures, which include declaration and management of interests. Specific individual conflict-of-interest details were not available in the supplied summaries, so no additional claims are made beyond WHO’s stated adherence to its handbook process [NCBI Bookshelf](#).

II. Evidence Summary After WHO Latest Guideline Release (Guideline on haemoglobin cutoffs to define anaemia in individuals and populations) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Post-guideline evidence was synthesized from the supplied summaries using targeted searches focused on:

- WHO 2024 haemoglobin cutoffs.
- Anaemia diagnosis and haemoglobin measurement.
- Oral versus intravenous iron therapy.
- Paediatric and adolescent iron-deficiency anaemia treatment.
- Pregnancy anaemia and iron supplementation.
- CKD-related anaemia guidance.
- WHO implementation and monitoring activities.

Representative queries included:

- WHO 2024 haemoglobin cutoffs anaemia children pregnancy altitude smoking
- anaemia management systematic review meta-analysis 2025
- oral intravenous iron anaemia systematic review 2025
- children adolescents iron deficiency anaemia dose duration meta-analysis 2025
- CKD anaemia KDIGO 2026 haemoglobin ESA iron guideline

Databases and Sources Searched

The supplied evidence included:

- WHO publication pages and WHO-hosted PDFs.
- NCBI Bookshelf WHO guideline records.
- Guideline Central summaries.
- PubMed-indexed systematic reviews and meta-analyses.
- PLOS One.
- Scientific Reports.
- BMJ/Gut guideline material.
- UK Kidney Association guideline.
- KDIGO 2026 summary via NephJC.
- WHO advisory-group reports.

Inclusion Criteria

Included evidence met at least one of the following criteria:

- Published or updated after **5 March 2024**, the WHO haemoglobin-cutoff guideline release date.
- Directly addressed anaemia diagnosis, haemoglobin cutoffs, measurement, prevention, treatment, or management.
- Came from WHO, other authoritative guideline bodies, systematic reviews, meta-analyses, or relevant clinical evidence sources.
- Reported outcomes relevant to haemoglobin, anaemia prevalence, iron deficiency, maternal/neonatal outcomes, transfusion, or treatment safety.

Exclusion Criteria

Excluded or down-weighted evidence included:

- Non-anaemia-specific youth-engagement material, except where needed to clarify the requested title.
- Sources without extractable anaemia-related methods or findings.
- Non-systematic opinion pieces unless used only for implementation context.
- Evidence predating the WHO 2024 guideline unless necessary as background.

2.2 Key New Studies and Evidence After 5 March 2024

Study or evidence source	Design	Key findings	GRADE Quality	Strength / Implication
WHO 2024 haemoglobin-cutoff guideline implementation summaries, including Guideline Central update	Guideline summary and secondary guideline dissemination	Confirms WHO 2024 cutoffs, including <105 g/L for children 6–23 months; trimester-specific pregnancy cutoffs; altitude and smoking adjustment; no adjustment for infection/inflammation Guideline Central .	Moderate	Strong for adopting WHO cutoffs
Peru-focused analysis of WHO 2024 guideline implications	Policy/implementation analysis	Highlights three practical changes: lower cutoff for children 6–23 months, revised altitude adjustment, and preference for venous blood measured using calibrated automated or portable haematology instruments under quality control SciELO PDF .	Low	Conditional implementation support

Study or evidence source	Design	Key findings	GRADE Quality	Strength / Implication
WHO-UNICEF TEAM 2025 haemoglobin measurement work	Evidence-review programme report	Cornell University was selected to synthesize evidence on haemoglobin measurement methods and blood sources; >10,000 articles screened, 44 included and 201 awaiting classification at the time of reporting WHO-UNICEF TEAM report .	Low pending final review	Conditional ; may inform future measurement recommendations
WHO STAGE 2025 report	WHO advisory-group implementation report	Implementation and monitoring guides were under development, including modules on direct causes and underlying risk factors of anaemia; drafts of six modules were expected by end of 2025 WHO STAGE report .	Low	Conditional implementation relevance
WHO STAGE 2025 anaemia advocacy and post-2030 agenda	WHO advisory-group report	WHO was advised to elevate anaemia as a priority, strengthen resource allocation, and consider post-2030 anaemia targets for infants, young children, and other high-risk groups WHO STAGE report .	Very Low for clinical effect; Moderate for policy relevance	Conditional

Study or evidence source	Design	Key findings	GRADE Quality	Strength / Implication
2025 systematic review/meta-analysis: oral versus intravenous iron for iron-deficiency anaemia	Systematic review and meta-analysis; 31 studies	IV iron was generally superior to oral iron for haemoglobin improvement in CKD, IBD, cancer-related anaemia, and general iron-deficiency anaemia; no clear advantage in postoperative patients or restless legs syndrome PubMed , Biomedical Reports .	Moderate	Conditional use of IV iron by clinical context
2025 paediatric/adolescent iron dose-duration meta-analysis	Systematic review and meta-analysis	Low-dose iron <5 mg/kg/day with treatment duration ❤️ months or >6 months was optimal for haemoglobin improvement in children/adolescents with IDA; treatment should be tailored by baseline Hb and severity PubMed , PLOS One .	Low to Moderate	Conditional; supports individualized paediatric treatment

Study or evidence source	Design	Key findings	GRADE Quality	Strength / Implication
2025–2026 CKD anaemia guidance summaries	Guideline and expert summary	UK Kidney Association reports CKD anaemia thresholds and paediatric CKD investigation thresholds; KDIGO 2026 summary recommends maintaining Hb <11.5 g/dL in adults receiving ESA therapy because higher targets increase stroke, vascular access thrombosis, and cardiovascular events UKKA PDF , NephJC .	Moderate for harm signal; Very Low for exact target certainty as reported	Strong caution against high ESA targets
2026 Scientific Reports cancer-related anaemia study	Prospective clinical study	Used WHO anaemia criteria of Hb <13.0 g/dL in men and <12.0 g/dL in women ; defined iron deficiency using ferritin and TSAT, including functional iron deficiency in inflammation Scientific Reports .	Low	Conditional diagnostic support

2.3 Diagnostic and Management Synthesis

2.3.1 Diagnosis of Anaemia

Use a staged diagnostic approach:

1. Confirm anaemia using haemoglobin

- Apply WHO 2024 age-, sex-, pregnancy-, and trimester-specific cutoffs.
- Adjust for altitude and smoking where applicable [Guideline Central](#).

2. Determine severity and urgency

- Severe anaemia thresholds from older WHO materials include Hb **<70 g/L** for pregnant women and young children, and **<80 g/L** for non-pregnant women, though current severity grading should be interpreted alongside local protocols and the 2024 diagnostic framework [WHO Iron Deficiency Anaemia PDF](#), [WHO strategies PDF](#).

3. Identify cause

- Iron deficiency is common but not universal.
- Evaluate ferritin, transferrin saturation, red cell indices, inflammation, malaria, helminths, chronic infection, kidney disease, haemoglobinopathies, B12/folate deficiency, and chronic disease where clinically indicated [WHO anaemia topic page](#).

4. Interpret iron biomarkers carefully

- Ferritin **<30 µg/L** supports iron deficiency when inflammation is absent.
- In inflammation, higher ferritin thresholds such as **<100 µg/L** may be used, with TSAT **<20%** supporting iron deficiency [Iron deficiency anaemia review](#).

GRADE Table: Diagnostic Approach

Diagnostic action	Evidence basis	GRADE Quality	Strength
Use WHO 2024 haemoglobin thresholds as the primary diagnostic entry point.	WHO 2024 guideline WHO, Guideline Central	Moderate	Strong
Adjust Hb for altitude and smoking.	WHO 2024 guideline summary Guideline Central	Moderate for altitude; Low for smoking	Strong for altitude; Conditional for smoking
Use ferritin/TSAT and inflammatory context to diagnose iron deficiency, not haemoglobin alone.	Practical review and policy-aligned evidence Iron deficiency anaemia review , Scientific Reports	Moderate	Strong
Prefer full blood count for pregnancy anaemia diagnosis where available.	WHO maternal guidance WHO consolidated guideline PDF	Moderate	Strong

2.4 Treatment and Prevention Synthesis

2.4.1 Pregnant Women

WHO recommends daily oral iron and folic acid:

- 30–60 mg elemental iron
- 400 µg folic acid
- Daily during pregnancy

Purpose: prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth [WHO ELENA](#), [WHO consolidated guideline PDF](#).

Intermittent pregnancy option:

- **120 mg elemental iron + 2.8 mg folic acid once weekly**
- Use when daily iron is not acceptable because of side effects and when anaemia prevalence among pregnant women is **<20%** [WHO WIFAS PDF](#).

IV iron in pregnancy:

- Use when oral iron cannot be used, is not tolerated, or rapid correction is clinically required, provided monitoring for anaphylaxis is available [WHO consolidated guideline PDF](#).

Pregnancy recommendation	GRADE Quality	Strength
Daily oral iron + folic acid for all pregnant women in relevant settings	Moderate	Strong
Weekly intermittent iron + folic acid when daily iron is not tolerated and population prevalence is <20%	Low	Conditional
IV iron for pregnancy IDA when oral iron is unsuitable or rapid correction is needed	Moderate	Conditional

2.4.2 Infants and Children

WHO’s daily iron supplementation guideline supports daily iron supplementation as a public-health intervention to prevent anaemia and iron deficiency in infants and children [WHO infants/children guideline page](#).

Reported evidence quality for daily oral iron in children aged **6–23 months** varied by outcome [WHO infants/children guideline PDF](#):

Outcome	WHO-reported evidence quality
Iron-deficiency anaemia	High
Anaemia	Moderate
Stunting	Moderate
Wasting	Low
Mortality	Low

A 2025 review summarizing WHO prophylactic guidance reported age-based preventive elemental iron doses of:

- **10–12.5 mg/day** for children aged **6–23 months**
- **30 mg/day** for children aged **24–59 months**
- **30–60 mg/day** for ages **5–19 years**
- Usually for **three consecutive months**, with daily or intermittent schedules depending on local anaemia prevalence [PLOS One](#).

Infant/child recommendation	GRADE Quality	Strength
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Infant/child recommendation	GRADE Quality	Strength
Daily iron supplementation to prevent iron-deficiency anaemia in appropriate high-risk child populations	High for IDA outcome; Moderate for anaemia	Strong
Tailor treatment dose/duration for paediatric IDA by baseline Hb and severity	Low to Moderate	Conditional

2.4.3 Menstruating Adolescent Girls and Women

WHO-linked guidance recommends intermittent iron and folic acid supplementation in menstruating adolescent girls and women in settings where anaemia prevalence among non-pregnant adolescent girls and women aged 15–49 years is $\geq 20\%$ [WHO WIFAS PDF](#).

Commonly cited regimen:

- **60 mg elemental iron**
- **2.8 mg folic acid**
- **Once weekly**

Purpose: improve haemoglobin concentration and iron status and reduce anaemia risk [WHO WIFAS PDF](#).

Recommendation	GRADE Quality	Strength
Weekly iron and folic acid for menstruating adolescent girls and women where anaemia prevalence is $\geq 20\%$	Moderate	Strong

2.4.4 Adults With Iron-Deficiency Anaemia

Post-2024 evidence supports individualized route selection:

- Oral iron remains appropriate for many stable patients with uncomplicated iron deficiency.
- IV iron is generally more effective for haemoglobin improvement in CKD, IBD, cancer-related anaemia, and general IDA, but not clearly superior in all contexts [PubMed](#), [Biomedical Reports](#).

Adult IDA treatment action	Evidence	GRADE Quality	Strength
Use oral iron for stable uncomplicated IDA when absorption and adherence are likely.	Established practice; comparative evidence supports effectiveness	Moderate	Strong
Use IV iron when oral iron is ineffective, not tolerated, malabsorption is likely, inflammation is present, rapid correction is needed, or CKD/IBD/cancer-related anaemia is present.	2025 systematic review/meta-analysis PubMed	Moderate	Conditional

Adult IDA treatment action	Evidence	GRADE Quality	Strength
Do not use haemoglobin alone to diagnose IDA; evaluate ferritin, TSAT, red-cell indices, and inflammatory context.	Diagnostic reviews Iron deficiency anaemia review	Moderate	Strong

2.4.5 Chronic Kidney Disease-Related Anaemia

CKD-specific anaemia management requires special caution:

- UK Kidney Association 2025 guidance reports CKD anaemia diagnostic thresholds aligned with WHO/KDIGO: Hb **<130 g/L** in males and **<120 g/L** in females [UKKA PDF](#).
- KDIGO 2026 summary recommends maintaining Hb **below 11.5 g/dL** in adults receiving ESA therapy, reflecting evidence that higher ESA targets increase stroke, vascular access thrombosis, and cardiovascular events [NephJC](#).

CKD anaemia action	GRADE Quality	Strength
Evaluate CKD patients for anaemia using sex-specific Hb thresholds and iron indices.	Moderate	Strong
Avoid high haemoglobin targets with ESA therapy; maintain Hb below 11.5 g/dL in adults on ESA therapy.	Moderate for harm direction; exact target certainty reported as very low in KDIGO summary	Strong
Consider IV iron when iron deficiency is present and oral iron is unlikely to be sufficient or effective.	Moderate	Conditional

2.5 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Priority
Full validation of WHO 2024 haemoglobin cutoffs in diverse ancestry, geographic, altitude, pregnancy, and age groups	Older cutoffs were based on limited populations; 2024 update improves this but further validation is needed	High
Optimal haemoglobin measurement method and blood source	WHO-UNICEF TEAM 2025 review found extensive literature but many exclusions; results may inform future recommendations WHO-UNICEF TEAM report	High
Best paediatric IDA treatment dose and duration	2025 meta-analysis suggests low-dose strategies may be optimal, but heterogeneity remains PubMed	High
Comparative effectiveness of oral versus IV iron by cause of anaemia	IV iron benefits vary by CKD, IBD, cancer, postoperative status, and inflammation	High

Evidence gap	Why it matters	Priority
Implementation strategies for high-burden settings	WHO STAGE reported implementation and monitoring guides under development WHO STAGE report	High
Integration of anaemia targets into post-2030 global nutrition agenda	WHO STAGE recommended considering anaemia targets for infants, young children, and other high-risk groups WHO STAGE report	Moderate to High
Meaningful youth engagement in anaemia guideline development	WHO youth-engagement case study exists for abortion care, but no anaemia-specific youth-engagement case study was identified WHO event page	Moderate

2.6 Updated Recommendations Based on New Evidence

Original WHO recommendation	New evidence impact after 5 March 2024	Updated recommendation as of 29 April 2026
Use WHO haemoglobin cutoffs to define anaemia.	Guideline summaries and implementation analyses reinforce the 2024 cutoffs, especially <105 g/L for children 6–23 months Guideline Central , SciELO PDF .	Adopt WHO 2024 cutoffs in clinical and public-health systems; ensure laboratories and surveys update child 6–23 month thresholds.
Adjust haemoglobin for altitude and smoking.	Peru implementation analysis highlights major effects of revised altitude adjustment SciELO PDF .	Continue altitude and smoking adjustment ; train programmes on the revised altitude equation and avoid using obsolete adjustment formulas.
Use haemoglobin testing for anaemia diagnosis.	WHO-UNICEF TEAM review on measurement methods was underway, with large evidence screening WHO-UNICEF TEAM report .	Prefer quality-controlled venous blood measurement with calibrated automated or portable haematology instruments where feasible; update when WHO releases new measurement recommendations.
Provide daily iron and folic acid in pregnancy.	No post-2024 evidence overturns WHO recommendation.	Maintain daily 30–60 mg elemental iron + 400 µg folic acid for pregnant women, with intermittent regimen only where criteria are met.
Use IV iron selectively in pregnancy and other IDA contexts.	2025 meta-analysis supports IV iron superiority for Hb improvement in CKD, IBD, cancer-related anaemia, and general IDA PubMed .	Use IV iron selectively when oral iron is unsuitable, ineffective, not tolerated, malabsorption/inflammation is likely, or rapid correction is needed.

Original WHO recommendation	New evidence impact after 5 March 2024	Updated recommendation as of 29 April 2026
Provide iron supplementation in infants and children in appropriate settings.	2025 paediatric/adolescent meta-analysis suggests low-dose treatment strategies may optimize Hb recovery PubMed .	Maintain WHO preventive supplementation; for treatment of paediatric IDA, individualize dose and duration by severity and response.
Manage CKD anaemia with iron and ESA therapy when indicated.	2025–2026 CKD guidance emphasizes avoiding high ESA Hb targets due to cardiovascular and thrombotic harms NephJC .	In CKD patients on ESA therapy, avoid normalization of Hb; maintain conservative Hb targets, generally <11.5 g/dL in adults on ESA therapy.

2.7 Actionable Clinical and Public-Health Algorithm

Step 1 — Screen and define anaemia

- Measure haemoglobin using reliable methods.
- Apply WHO 2024 cutoffs by age, sex, pregnancy trimester, and population group.
- Adjust for altitude and smoking.

Step 2 — Assess severity and urgency

Urgent evaluation is required for:

- Severe anaemia.
- Symptoms of hypoxia, syncope, heart failure, angina, severe fatigue, or tachycardia.
- Pregnancy with moderate/severe anaemia.
- Infants, young children, older adults, CKD patients, cancer patients, and people with active bleeding.

Step 3 — Determine cause

Minimum workup where available:

- Full blood count and red-cell indices.
- Ferritin.
- Transferrin saturation.
- C-reactive protein or inflammation marker if ferritin interpretation is uncertain.
- Reticulocyte count when haemolysis or marrow response is relevant.
- B12, folate, renal function, malaria testing, stool/parasite evaluation, haemoglobinopathy testing, or gastrointestinal evaluation depending on context.

Step 4 — Treat cause-specific anaemia

Cause or context	Preferred management
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Cause or context	Preferred management
Nutritional iron deficiency, stable patient	Oral iron; dietary counselling; investigate blood loss
Pregnancy prevention	Daily iron + folic acid
Pregnancy IDA with intolerance, malabsorption, severe anaemia, or need for rapid correction	IV iron with monitoring
Children in high-risk settings	WHO preventive iron supplementation; treat confirmed IDA with individualized dosing
Menstruating adolescents/women in ≥20% prevalence settings	Weekly iron + folic acid
CKD anaemia	Iron assessment/repletion, ESA if indicated, avoid high Hb targets
Inflammatory disease, IBD, cancer, CKD	Consider IV iron when oral iron is ineffective or inappropriate
Haemoglobinopathy, B12/folate deficiency, malaria, helminths, chronic infection	Cause-specific therapy; avoid assuming all anaemia is iron deficiency

Step 5 — Monitor response

- Recheck haemoglobin after an appropriate treatment interval.
- For oral iron, assess adherence, tolerance, ongoing bleeding, malabsorption, inflammation, and diagnosis if response is inadequate.
- Continue iron after haemoglobin correction long enough to replenish iron stores, according to local protocols and patient context.
- Avoid excessive iron in patients without confirmed deficiency or with iron-loading disorders.

2.8 Bottom Line

As of **29 April 2026**, the **WHO 2024 haemoglobin-cutoff guideline** remains the most recent WHO anaemia-specific diagnostic guideline. The key change is the lower anaemia cutoff for **children aged 6–23 months: haemoglobin <105 g/L**. WHO recommendations for anaemia prevention and management continue to emphasize accurate haemoglobin measurement, adjustment for altitude and smoking, cause-specific diagnosis, iron and folic acid supplementation in pregnancy, preventive iron strategies for infants/children and menstruating adolescent girls/women in high-prevalence settings, and selective use of IV iron when oral iron is unsuitable or rapid correction is needed. Post-2024 evidence strengthens the case for individualized iron route selection and highlights the need for better evidence on haemoglobin measurement methods, paediatric treatment dosing, and implementation in high-burden populations.